

SnapGuardian AI

Zero-Cloud Privacy, Task Context and NPU Thermal Engine
Designed for Snapdragon-Powered HP PCs

Problem Statement

- Privacy Risks: Developers and enterprise users expose API keys, stack traces, and documents to cloud services.
Device Overheating: Heavy AI multitasking heats CPU cores to 82C, causing severe thermal throttling and battery drain.

SnapGuardian AI Solution

- SnapCooler NPU Offloader: Offloads AI workloads to 45 TOPS Hexagon NPU (1.2W TDP), dropping CPU temps by 32C.
Real-Time Vision Privacy Guard: Obfuscates secrets locally on screen frames.
Task Context Graph: 100% offline graph mapping for proactive error root cause analysis.

Empirical Benchmarks and Results

- Latency: 1.82 ms (NPU) vs 14.5 ms (CPU) - 7.97x Speedup
Power Consumption: 1.2 W (NPU) vs 8.5 W (CPU) - 85.8% Energy Savings
Thermal Delta: -32C CPU Surface Temperature Reduction
Network Traffic: 0 Network Requests (100% Offline Privacy)